Lab I - BevOS: A Fresh Take on Enterprise Resource Reasoning

Julien P. Fernandes

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Professor Thomas J. Kennedy

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1. Introduction

1.1 Societal Problem

The necessity to coordinate increasingly globally distributed manufacturing networks over the latter half of the 20th century led to centralized, software-based planning through the 1990s and 2000s. Today the beverage manufacturing industry generates billions of dollars annually and supports thousands of companies throughout the United States with the beverage packaging market alone valued at approximately $157.7 billion in 2023 and is projected to reach $222.1 billion by 2030 (Grand View Research, 2024).

Despite decades of advances in both manufacturing and software management technology fragmented, legacy processes consisting of single copy data files, email communication chains, and tribal knowledge constrained to human bureaucracy is a continued weight on business. These inefficiencies create communication delays, increase operational costs, reduce visibility into production activities, and limit organizational scalability (Research and Markets, 2025). The complexity of beverage manufacturing and packaging investments set high bars to clear in terms of the need for coordinated, scalable information systems that help organizations manage production activity across multiple stakeholders (Zimmerman, 2026).

1.2 Problem and Needed Solution Characteristics

Overwhelmingly, the large adoption cost and complexity associated with moving to a unified operational infrastructure managed through software remains the defining challenge of Enterprise Resource Management and Reasoning in the modern day. The rise of Enterprise Resource Planning (ERP) and Manufacturing Execution Systems (MES) saw early successes but also highlighted integration difficulties arising from disconnected proprietary systems.

Thus, in considering a ‘Fresh Take’ in this problem space we first aim to preserve the gains and lessons from ERP and MES software. An effective solution must provide centralized workflow coordination, real-time visibility into manufacturing activities, automated approval processes, document management, production tracking, and stakeholder communication. The system must also maintain a single source of truth, support data-driven decision-making, and provide scalable architecture capable of adapting to future business growth. However, to move beyond we will further consider how software design can work to minimize adoption cost and proprietary system integration cost.

1.3 Product Introduction

BevOS is a centralized workflow management platform designed specifically for beverage manufacturers as proof of concept along the development path of an Operational Infrastructure as a Service (OlaaS) platform that bridges enterprise resource reasoning with a philosophy centered on minimizing adoption friction and working with disconnected systems where needed. The system connects brands, suppliers, distributors, co-packers, logistics providers, and internal operations through a single platform that centralizes communication, workflow management, production tracking, logistics coordination, and business documentation.

Rather than necessitating itself as the sole source of truth for all operations, BevOS acts as a best-try representation of truth on the ground as understood by the software platform, thus BevOS works within existing business processes and paradigms as much as possible while also working to improve efficiency by reducing manual coordination, improving communication, and increasing operational visibility. While bringing adoption costs and system disruption to nothing is understood to be impossible a functional prototype will demonstrate how the product's interface, services, modules, database, and development philosophy work together to address the identified coordination problem by learning with a domain industry instead of mandating it.

2. BevOS Product Description

BevOS is a centralized workflow management platform designed specifically for beverage manufacturing operations. The platform streamlines communication, production scheduling, approval tracking, logistics coordination, and document management by bringing key stakeholders into a single system. BevOS aims to improve operational efficiency, reduce manual processes, increase visibility throughout the production lifecycle, and support scalable business growth.

2.1 Purpose

BevOS aims to centralize beverage manufacturing including workflow and operationally related information. As a central source of truth and tracing analysis the user will see a number of benefits including a reduction in dependence on spreadsheets, email chains, phone calls, and otherwise fragmented applications and data stores. This allows the consistent coordination of multiple interest parties such as brands, distributors, suppliers, production personnel, logistics providers, and finance team operating from a concurrent base of information and underlying industry infrastructure. Ultimately the system aims to provide a shard operational view without attempting to insist on the replacement of every existing enterprise system or human decision-making process.

2.2 Goals and Objectives

To accomplish this purpose, BevOS establishes the following objectives as guides on platform design and primary operational improvement. Overall improvement in coordination, visibility, and scalability of beverage manufacturing operations should include,

- Improve operational efficiency by reducing repeated manual coordination.
- Reduce communication and approval delays.
- Increase visibility throughout the production lifestyle.
- Organize documents and operational records within their related workflows.
- Support timely, data-informed management decisions through reports and dashboards.

- Provide a modular foundation that can adapt as operational processes and business need evolve.



Figure 1. BevOS Objective Breakdown

2.3 Intended Users

Brands initiate product requests and monitor progress as their overall product market through the system moves from inception to active sales and recurring production orders. Distributors coordinate product availability according to their own stores and coordinate logistics to fulfill order invoicing as demand and supply change. Suppliers receive quote requests and provide estimates alongside support in material or service needs. Various authorized personnel as operational employees of these industrial partners or another coordinating party therein access webapp or other user areas to contribute information or otherwise assess workflows.

2.1. Key Product Features and Capabilities

2.1.1. Workflow Management

BevOS centralizes operational activities and tracks work throughout the production lifecycle. Requests, approvals, documents, communications, production activities, and logistics information is all connected within coordinated workflows reducing fragmentation across platforms and keeping related information and activities together through the manufacturing process.

2.1.2. Product Request Management

BevOS allows authorized users to create, review, and monitor product requests within a consistent workflow environment. Request details and status information are updated as needed throughout reducing the risk of incomplete requests, or otherwise lost information during initial stages of deal/production order creation.

2.1.3. Quote Management

BevOS creates and tracks supplier quote requests, responses, and other related supplier resources. Quote information remains associated with the request that generated its query improving up to date supplier selection and cost evaluation while reducing the need for repeated follow-up where possible.

2.1.4. Approval Tracking

BevOS manages artwork, purchase orders, and other approval items needed throughout workflows by keeping record of their status, responsible parties, and history of interactions. Notifications and escalations support the process by providing human intervention and visibility where needed on outstanding actions.

2.1.5. Production Management

BevOS coordinates production schedules, milestones, and other real-time status information while connecting to subsequent manufacturing activities or partners in one dashboard. Emphasis is placed on allowing teams to identify delays or patterns as early as possible.

2.1.6. Logistics Coordination

BevOS tracks shipping activity and delivery status while integrating information from distribution and logistics activities. Emphasis on providing stakeholders across the product line with a clearer view of where a product is in the fulfillment process, especially in traditionally more opaque areas.

2.1.7. Document Management

BevOS stores and organizes artwork, purchase orders, production files, shipping records, and other operational documents in an archive manner for later access by users or organizations as needed. Keeping versioning information with specific files and communications helps prevent separation or loss of concern throughout the process.

2.1.8. Communication Platform

BevOS centralizes communication and updates among authorized stakeholders in as centralized a platform as possible trying to avoid ignoring incessant alarms or so on. Communications are also associated with relevant requests, quotes, approvals, production, etc. where possible.

2.1.9. Reporting and Dashboards

BevOS summarizes important information through reports and dashboard tailored to the production domain in question. Role-appropriate views provide stakeholders with relevant information for monitoring and their own executive decision making. Emphasis on taking stored data and turning it into useful perspectives or pattern recognition.

2.1.10. Issue Tracking

BevOS records bugs and problems automatically alongside provider user areas for feedback and ticket creation along with recording of responsible parties, status, and resolution information. Emphasis on using issue tracking and resolution to identify process weaknesses and inform project development overall.

2.2. Major Components (Hardware/Software)

2.2.1. Hardware Components

BevOS is envisioned as almost entirely software with hardware consideration typically in the cases where the domain / manufacturing industry, etc. would require such consideration for compatibility with existing systems. Users access BevOS primarily from internet browser capable devices through webapp frontend interfaces while backend work is handled on workstation / web server hardware.

2.2.2. Software Components

BevOS is composed of modular software components coming together to represent and support operational workflows. Broadly these components separate into user interaction/services, shared application services, specialized domain/operation functions, and persistent data storage allowing data to flow between each area in a common system architecture. See Figure 2 for a prospective design for BevOS graphical user interface.



Figure 2. BevOS GUI Mockup

- BevOS Web User Interface: provides the user-facing interface through which brands, distributors, and suppliers access authorized system functions.
- BevOS Core Services: receive requests from UI, applies workflow and access rules, coordinates functional modules, and manages data exchange.
- Quotes: manage quote requests, supplier responses, and related quote information.
- Communications: organizes messages, notifications, and workflow-related updates.
- Production: manages production schedules, milestones, status, and progress information.
- Logistics: manages shipment coordination and delivery-status information.
- Document storage: stores and retrieves operational files associated with BevOS workflows.
- Reporting: presents operational summaries, dashboards, and decision-support information.
- PostgreSQL: stores structured application data, including users, requests, quotes, approvals, production records, logistics records, and report data.

2.2.3. MFCD Structure and Data flow.

User activity enters through the BevOS interface and is coordinated by Core Services, which connects the platform's specialized functional modules with shared data and storage resources. This structure separates system responsibilities while maintaining a common flow of operational information across the platform.

- Brands, distributors, and suppliers interact with the system through BevOS UI.
- The BevOS UI sends user requests to BevOS Core Services.
- BevOS Core Services coordinates the production, logistics, quotes, communications, reporting, and document storage.
- The functional modules use PostgreSQL to store and retrieve the structured information needed to complete their responsibilities.
- This modular structure separates the user interaction, shared application logic, specialized operational functions, and persistent data storage.
- The separation supports maintainability and scalability because modules can evolve while continuing to use common core services and data.

3. Glossary

- **Approval Workflow** - A process used to review and approve operational tasks before they move to the next stage.
- **BevOS** - A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services** - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram)** - A diagram illustrating the primary hardware and software components of the system.
- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL** - An open-source relational database management system used to store application data.
- **Role-Based Access Control** - A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth** - A trusted, centralized set of current information used consistently by authorized stakeholders.
- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow** - A sequence of business activities completed to achieve a specific objective.

4. References

Grand View Research. (2024). *Beverage Packaging Market Size, Share & Trends Analysis Report.*

Honnappa, K. (2026). *Platform Visual Aid*. Nice Package Co.

Kennedy, T. (2026). *CS410 Professional Workforce Development.*

Morley, M. (2015, February 22). *The evolution of the digital manufacturing business*.

Wilson, T. (2026). *BevOS Presentation / Nice Package Co. Guest Presentation.*

Zimmerman, S. (2026) *How the food and beverage industry is rethinking manufacturing investments. Food Dive.*

Morley, M. (2015, February 22). *The evolution of the digital manufacturing business*. *Mark Morley*.